Direct Reactions With FRESCO; a Practical Guide

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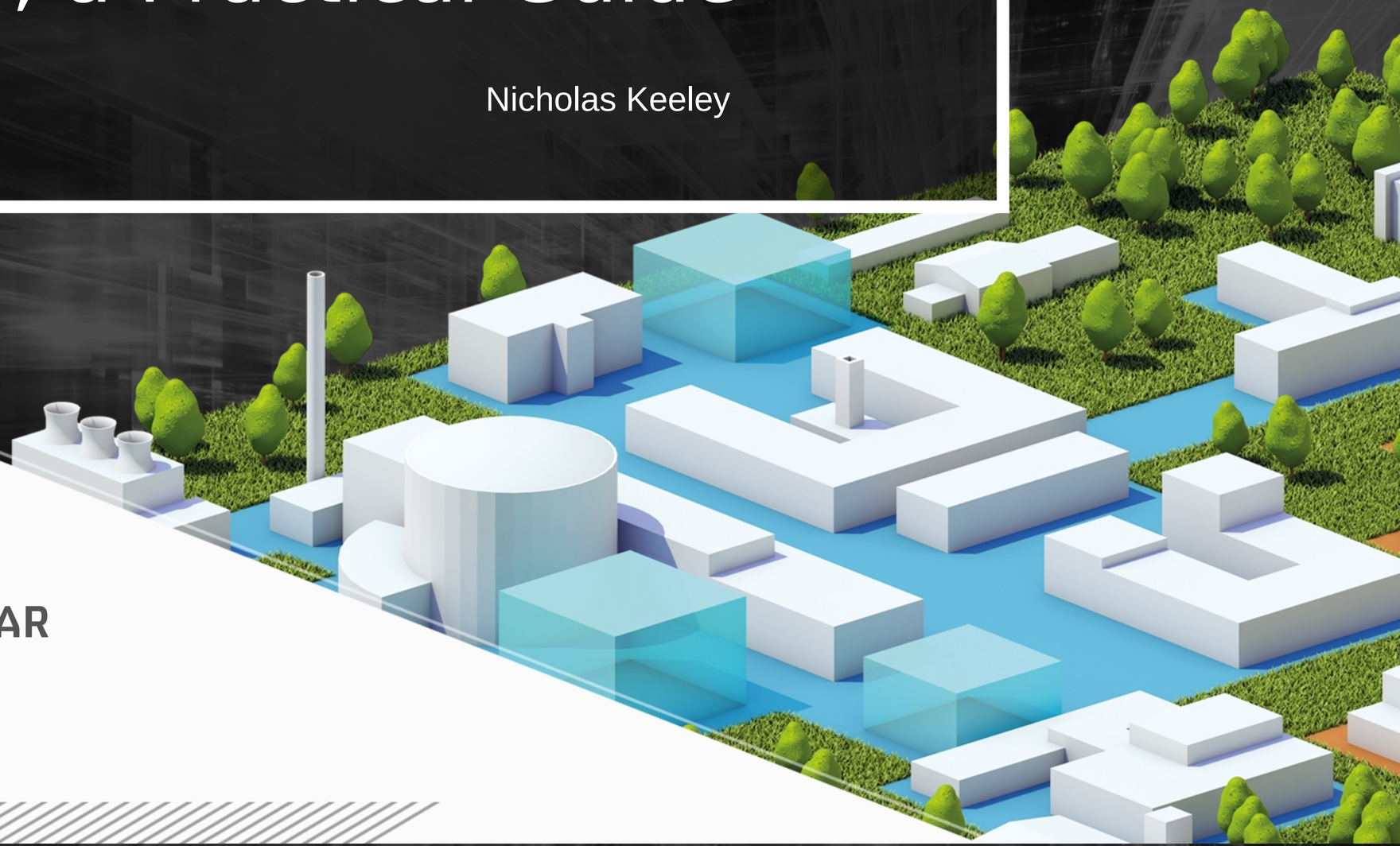

These lectures and the accompanying exercises will cover the practical aspects of performing direct reaction calculations using the code FRESCO, written by Ian Thompson; Comput. Phys. Rep. **7**, 167 (1988)

They will cover the optical model (including the use of SFRESCO for parameter searches), coupled channels for inelastic scattering, and distorted wave Born approximation (DWBA) and coupled reaction channels (CRC) calculations for transfer reactions

We will explore how to assemble the appropriate input files and the points to be aware of (and traps for the unwary) as well as some of the useful quantities that FRESCO can output for us

The Optical Model

We begin at the beginning, the simplest reaction process described by the simplest model: the optical model of elastic scattering.

You should by now be aware of the theoretical background of the model and the ambiguities inherent in fitting data with empirical optical potential forms (the parameters obtained by a given search are by no means unique)

In practical terms, optical model calculations fall naturally into two groups: those involving nucleons and light ions (d, t, ^{3}He, ^{4}He) as projectiles and those involving heavy ion scattering

We begin with (almost) the simplest case, proton elastic scattering …

A sample input file: 49.5 MeV protons incident on a ^{12}C target

```
Card 0    12C(p,p)12C at 49.5 MeV
Card 1    0.10    15.0    0.10
Card 2     00. 20.  +.00   F F
Card 3    1  0.0    -180.   1.0  1
Card 4    0.0    0   0 1 0  48
Card 5     1 1 0 0 1 3 0 0 0 0 0 0 0
Card 6    1H      1.0078  1.0        1  12C     12.0000 6.0       +0.0000
Card 7    0.5   +1 0.0                1  0.0   +1 0.00

Card 9      1 0  0    12.0     0.0    1.24
Card 10     1 1  0    35.7    1.15    0.67    5.84    1.25    0.80
Card 10     1 3  0    4.62    0.62    0.26
          0

Card 18     0   1   1
Card 19   49.5
          EOF
```

Card numbers are **card type**, as per FRESCO manual. Cards labelled in red represent “control” parameters and **cards labelled in bold will seldom (if ever) require changing.**

I will now give verbal explanations of what the various parameters do and how they are chosen.

How do we run FRESCO using this input file (or any input file, for that matter …) and what are the output files?

To run FRESCO, assuming you have correctly installed it and included the path to the executable file in your set-up, simply type the following command (from the directory containing your input file):

prompt:fresco<inputfilename>outputfilename &

The files that immediately concern us are the large one, called here “outputfilename”, and fort.201, which contains the elastic scattering angular distribution in the centre of mass frame, and fort.16, which contains the angular distributions for all channels included in the input file. In this particular case fort.201 and fort.16 are identical.

What do these files look like? Taking fort.201 first:

```
# 181 angles,  2 tensor ranks for 0=projectile
@subtitle   "12C(p,p)12C at 49.6 MeV"
@legend ON
@legend x1 0.2
@legend y1 0.8
@g0 type LOGY
@yaxis ticklabel format power
@legend string   0 "Lab energy =   49.6000"
#legend string   0 "Partition=  1 Excit=  1 near/far= 1"
#  Theta       sigma       iT11        T20         T21         T22         Kyy for proje
  0.1000E-01  0.3842E+15 -0.6814E-11
   1.000      0.3689E+07  0.4352E-04
   2.000      0.2049E+06  0.4002E-03
   3.000      0.3346E+05  0.1529E-02
   4.000       8225.      0.4087E-02
   5.000       2579.      0.8458E-02
```

Can be used in **xmgrace** to plot calculated angular distributions of differential cross section and analysing power(s). The main output file (“outputfilename” in our example …) contains much additional useful information:

```
FRESCO - FRES 3.2: Coupled Reaction Channels

12C(p,p)12C at 49.6 MeV

  0.100  15.000   0.100   0.000   0.000   0.000   0.000   0.000   0.000   0.000

Centre-of-mass Range is    152 * 0.1000 fm., Maximum at  15.00 fm., Interpolating NL forms every  0.10 fm.
Non - locality width is      4 * 0.1000 fm., Maximum of   0.30 fm., Centred at  0.00 fm.
2-Nucleon Separation of      0 * 0.1000 fm., Maximum of   0.00 fm., Minimum at  0.00 fm.
Maximum single particle bins of 15.0000 fm.
              M,Mint =          151         151


Range of total J is     0.0 <= J <=    20.0 (at least     0.0) and Absorbtion =>     0.0000 mb.
Dry Run = F, CC set limits =  0  0, Relativistic kinematics =  , Both/Far/Near Analyses =  1

Cross Sections (and up to T3 for 0=projectile) for Theta from   0.0 to-180.0 in steps of  1.0 degrees, DGAM=0,  Coordinates = 1 (Ma

Lower Radial Cutoff = maximum of -1.60*L*h &       0.0 fm.,  Lower Cutoff for Couplings =   0.0  fm.


Iterate Couplings between   0 and   0 times, to  0.000 % if sooner.
 Block solved exactly =   1 chs., with Pade = 0 & Isocen =  =0, NOSOL = F, initwf =  0
 NL quadrature with   48 Gaussian points, Calculate multipoles up to    26 from   0
 M-transfers for lp+lt greater than or equal to  6, Angular Integration Cutoff below  0.3906 %


Trace switches are :  CHANS =  1, LISTCC =  1, TRENEG =  0, CDETR =  0, SMATS =  1, XSTABL =  3, NLPL =  0

                      WAVES =  0, LAMPL =  0, VEFF =  0, KFUS =  0, WDISK =  0, BPM =  0, MELFIL =  0

                      CDCC =  0, NFUS =  0

Using unit mass = 1.000000 amu and 1/fine-structure constant = 137.03599 ( 1.000000 * true ) with hc = 197.32705 MeV.fm,
 thus 2*amu/hbar^2 = 0.0478450 = 1/20.9008 and Coulomb constant = 0.1574855 so e^2 = 1.43997, and nuclear magneton= 0.1261183

Now pre-scan input and save to file 3
```

```
********** PARTITION NUMBER  1 ***********************************************************************************

  PROJ=1H         MASS=  1.0078 Z=   1.0,        # STATES=    1T, TARG=12C      MASS= 12.0000 Z=   6.0,     Q-VALUE =   0.0000 MEV.


    1:  J= 0.5+ (B# 1), E=  0.0000, K= 0.5             Potl#  1       J= 0.0+ (B# 1), E=  0.0000, K= 0.0

 ********************************************************************************************************************************

 The following POTENTIALS are defined :

  KP#       TYPE      IT    SHAPE      at     V1        r1        a1        V2        r2        a2        A-in     A-used

                                          A#1       A#2       r0c       ac                                             h @ 1
    1    0=Coulomb       0=CHARGE (WS)  1    12.000     0.000    1.2400    0.0000    0.0000    0.0000     0.000    12.000   0.10000

    1    1=Volume        0=Woods-Saxon  2   35.7000    1.1500    0.6700    5.8400    1.2500    0.8000     0.000    12.000

    1    3=Projtl S.O.   0=Woods-Saxon  3    4.6200    0.6200    0.2600    0.0000    0.0000    0.0000     0.000    12.000

 Incoming partition  1 in excitation state # 1,   Laboratory Energy given for partition  1 Nucleus 1 in Excitation pair 1

0Lab. ENERGY ranges :

   from    49.6000     to   0.00000     in     0 intervals
   from    0.00000     to   0.00000     in     0 intervals
   from    0.00000     to   0.00000     in     0 intervals

 Largest real,imaginary parts of any form factor at R=   14.00 are   1.03E-01  5.25E-06 MeV
 Finished all Couplings @    4.38999850E-04           3

 Symmetric Hamiltonian
1********************************************************************************************************************************
********************************************************************************************************************************

        INCOMING 1H       ;    LABORATORY 1H       ENERGY =   49.600     MeV.

 ********************************************************************************************************************************
 ********************************************************************************************************************************

 Allocate arrays for            2  channels, of which            1  need wfs.

 ##############################################################################################################################
 #                                                                                                                            #
 # Total SPIN and PARITY =     0.5 +,      1 channels,    1 in 1st block.  Rmin & Coul turning =    0.0    0.2 fm.  #
 #                                                                                                                            #
 ##############################################################################################################################
```

```
   C Projectl Target   #   EX.   (L  Proj) J  + Targ = Jtotal     E-cm      Re K     Re Eta     RM*K            CH
   1 1H       12C      #    1: I   0 0.5    0.5  0.0     0.5    45.75716   1.42667   0.13469   21.4001  -0.46251  -0.19409 1 1 1
Reaction Xsec     0.5+/ 0 @ 1 =   8.985 , Out:   0.000#

Total SPIN, PARITY =     0.5 -,      1 chs,      1 cc.  Rmin & Coul turning =    0.0    0.2
Reaction Xsec     0.5-/ 1 @ 1 =   8.852 , Out:   0.000#

Total SPIN, PARITY =     1.5 +,      1 chs,      1 cc.  Rmin & Coul turning =    0.1    1.1
Reaction Xsec     1.5+/ 2 @ 1 =  16.119 , Out:   0.000#

Total SPIN, PARITY =     1.5 -,      1 chs,      1 cc.  Rmin & Coul turning =    0.1    1.1
Reaction Xsec     1.5-/ 1 @ 1 =  17.426 , Out:   0.000#

Total SPIN, PARITY =     2.5 +,      1 chs,      1 cc.  Rmin & Coul turning =    0.3    1.8
Reaction Xsec     2.5+/ 2 @ 1 =  25.885 , Out:   0.000#

Total SPIN, PARITY =     2.5 -,      1 chs,      1 cc.  Rmin & Coul turning =    0.3    1.8
Reaction Xsec     2.5-/ 3 @ 1 =  23.066 , Out:   0.000#

Total SPIN, PARITY =     3.5 +,      1 chs,      1 cc.  Rmin & Coul turning =    0.4    2.5
Reaction Xsec     3.5+/ 4 @ 1 =  23.877 , Out:   0.000#

Total SPIN, PARITY =     3.5 -,      1 chs,      1 cc.  Rmin & Coul turning =    0.4    2.5
Reaction Xsec     3.5-/ 3 @ 1 =  33.430 , Out:   0.000#

Total SPIN, PARITY =     4.5 +,      1 chs,      1 cc.  Rmin & Coul turning =    0.6    3.2
Reaction Xsec     4.5+/ 4 @ 1 =  31.086 , Out:   0.000#

Total SPIN, PARITY =     4.5 -,      1 chs,      1 cc.  Rmin & Coul turning =    0.6    3.2
Reaction Xsec     4.5-/ 5 @ 1 =  16.729 , Out:   0.000#

Total SPIN, PARITY =     5.5 +,      1 chs,      1 cc.  Rmin & Coul turning =    0.8    3.9
Reaction Xsec     5.5+/ 6 @ 1 =   9.624 , Out:   0.000#

Total SPIN, PARITY =     5.5 -,      1 chs,      1 cc.  Rmin & Coul turning =    0.8    3.9
Reaction Xsec     5.5-/ 5 @ 1 =  20.264 , Out:   0.000#
```

```
 Total SPIN, PARITY =    20.5 +,      1 chs,      1 cc.  Rmin & Coul turning =    3.2   14.5
 Reaction Xsec    20.5+/20 @ 1 =   0.247/, Out:   0.000#

 Total SPIN, PARITY =    20.5 -,      1 chs,      1 cc.  Rmin & Coul turning =    3.2   14.5
 Reaction Xsec    20.5-/21 @ 1 =   0.077/, Out:   0.000#
 Finished all CC sets @    1.71600003E-03
0CUMULATIVE REACTION cross section                =  268.90013  <L> =     3.77  <L**2> =     18.3
0CUMULATIVE outgoing cross sections in partition 1 :    0.00000

0Cumulative ABSORBTION by Imaginary Potentials     =  268.90013  <L> =     3.77  <L**2> =     18.3
   Fusion for specific 1H       M-states :  268.900126  268.900126
0CUMULATIVE OUTGOING cross section                =    0.00000
 To convert to S-factors (MeV.mb = keV.b),  multiply by  1.0666E+02


 CROSS SECTIONS FOR OUTGOING 1H       & 12C      in state #   1 with spins & parities 0.5  + &  0.0  +;  0

     0.01 deg.: X-S =   3.8422E+14 mb/sr,                    & pols = 0. -0.00000
+                                    /R =   1.0000E+00

     1.00 deg.: X-S =   3.6892E+06 mb/sr,                    & pols = 0.  0.00004
+                                    /R =   9.6014E-01

     2.00 deg.: X-S =   2.0493E+05 mb/sr,                    & pols = 0.  0.00040
+                                    /R =   8.5321E-01

     3.00 deg.: X-S =   3.3457E+04 mb/sr,                    & pols =-0.  0.00153
+                                    /R =   7.0500E-01

     4.00 deg.: X-S =  8225.298463 mb/sr,                    & pols =-0.  0.00409
+                                    /R =   5.4760E-01
```

```
   179.00 deg.: X-S =     0.015767 mb/sr,                       & pols =-0. -0.01999
+                                        /R =   7.0746E-01

   179.99 deg.: X-S =     0.015780 mb/sr,                       & pols =-0. -0.00020
+                                        /R =   7.0815E-01

                                         Integrated********** mb, over [   0.000, 180.000] at    49.6000 MeV
 Finished all xsecs @    3.51900002E-03

  The following files have been created:
   3:local copy of User input.            6:standard output.
  13:total cross sections/state.         16:tables of cross sections.
  39:cross sections for each Ecm.        40:all cross sectns. for each Elab.
  56:Fusion for each Jtotal.            201:Separate cross sections.

 PARAMETERS : MAXQRN   MFNL   MLOC  LMAX1 MCLIST
    ALLOWED :      1      1      3     40      5
    REQUIRED:      0      0      3     22      3


 ACCURACY ANALYSIS at    49.600 MeV :

  Elastic h*k = 0.143 so OK       compared with 0.200

   Real(S-el) > 0.01 first at J =       2.5
   Real(S-el) > 0.10 first at J =       3.5
   Real(S-el) > 0.50 first at J =       3.5

   Real(S-el) > 0.90 first at J =       5.5
   Real(S-el) > 0.99 first at J =       8.5
   R-turn =    14.46 fm    at J =      20.5

 Forward-angle excitation cut off below   0.735 deg by max JT =      20.5
                            and below   0.721 deg by max R.
```

Some points to be aware of with regard to the input:

The example uses Woods-Saxon potentials, but several other forms are available as standard inputs in FRESCO

Check the definition of the potential when using Woods-Saxon derivative forms; some codes (e.g. DWUCK) do not have the factor of 4

Fresco's definition of the spin-orbit is not completely standard; for some spin values you may need to divide by a factor of 2 (check carefully whether $\boldsymbol{\ell}\cdot\mathbf{s}$ or $\boldsymbol{\ell}\cdot\boldsymbol{\sigma}$ is used, where $\boldsymbol{\sigma} = 2\,\mathbf{s}$)

Any form of local potential may be calculated externally and read in on stream 4 (file fort.4)

The example is for proton scattering from a light target (^{12}C) but the same format is used for any projectile/target combination at any incident energy

Points to watch are:

Ensure you have allowed an adequate number of partial waves; in general higher energies and/or heavier systems will require more partial waves

Ensure the matching radius (RMATCH on card 1) is large enough; a useful rule of thumb is to take RMATCH equal to the classical turning point for L_{max}: $L_{max}(L_{max}+1) = k\times RMATCH\ (k\times RMATCH - 2n)$
where $k = 0.2187[m(u)E(MeV)]^{1/2}\ fm^{-1}$ and $n = 0.1575\ Z_1Z_2[M(u)/E(MeV)]^{1/2}$

Radial step size (HCM) should be small enough; 0.1 fm is a good starting point – FRESCO will suggest a value if this is too large

How do we know whether we have enough partial waves? This is a somewhat subjective question, with no “correct” answer as such. The value depends very much on the system and the incident energy and may vary from 10-15 for proton scattering to several hundred for some heavy ion scattering systems.

One way to check is to look at the file fort.56:

For OM calculations the output is particularly simple:

Column 1: J in units of ħ (we have non-zero channel spin here)
Column 2: absorption by imaginary potential in mb
Column 3: reaction cross section in mb (sum of columns 2 and 4)
Column 4: outgoing cross section in mb

```
# Fusion cross sections for ELAB=  49.600, ECM=  45.757
     0.5   17.84        17.84        0.000
     1.5   33.55        33.55        0.000
     2.5   48.95        48.95        0.000
     3.5   57.31        57.31        0.000
     4.5   47.81        47.81        0.000
     5.5   29.89        29.89        0.000
     6.5   16.40        16.40        0.000
     7.5   8.565        8.565        0.000
     8.5   4.357        4.357        0.000
     9.5   2.175        2.175        0.000
    10.5   1.069        1.069        0.000
    11.5  0.5192       0.5192        0.000
    12.5  0.2495       0.2495        0.000
    13.5  0.1187       0.1187        0.000
    14.5  0.5610E-01   0.5610E-01    0.000
    15.5  0.2633E-01   0.2633E-01    0.000
    16.5  0.1220E-01   0.1220E-01    0.000
    17.5  0.5475E-02   0.5475E-02    0.000
    18.5  0.2323E-02   0.2323E-02    0.000
    19.5  0.9099E-03   0.9099E-03    0.000
    20.5  0.3240E-03   0.3240E-03    0.000
 &
```

Any value < about 10^{-2} mb at L_{max} is usually fine for optical model calculations (and many other types, depending on requirements)

FRESCO calculates the turning point for L_{max} so that one can easily check whether RMATCH is large enough

Finally, the parameter RINTP on card 1. This is always a multiple of HCM, the integration step size. In OM calculations you may simply leave it the same value as HCM since it is not used here. It can be useful to make it larger in more complicated calculations (to speed things up) but care is needed …

You should now be equipped to assemble (in actuality modify) an optical model input file for FRESCO for any desired system

The practical exercises will take you through the steps involved in converting a sample input file to suit various situations to enable you to get an idea of the acceptable ranges of the various parameters

This is all very well for “one shot” calculations, but much of the time you will wish to adjust the parameters of the potential to fit a given elastic scattering data set.

We will now see how this can be done using SFRESCO, part of the FRESCO package (indeed, SFRESCO enables searching with any calculation that FRESCO can do – CC, DWBA, CRC etc. – and on a variety of variables)

To run SFRESCO two additional files are needed as well as the usual input file for FRESCO itself. In effect, SFRESCO uses FRESCO as a “sub-routine” with the MINUIT χ^2 minimisation package

One file contains a list of commands to pass to SFRESCO (ultimately MINUIT) and one controls which variables are to be searched on and contains the data set(s) that are to be fitted

We shall begin with the “control file” which we will call sf.inp – for most purposes this will not need to be changed, although those familiar with MINUIT may wish to use different minimisation techniques etc.

`search.in`	Search file name
`min`	Call minuit
`migrad`	Minuit command; minimise using MIGrad
`end`	Return to SFRESCO
`q`	Query status of search variables (print values)
`show`	List all data sets and predictions with χ^2 values
`plot`	Write file search.plot with data plus fit curve

We now look at the file containing the list of variables to be searched on and the data set(s) we wish to fit. For this first example we shall only fit one data set, the angular distribution of the elastic scattering differential cross section

For elastic scattering we could in principle fit analysing power data as well as (or instead of) the cross section. For coupled channels or DWBA/CRC we could also fit inelastic scattering and/or transfer angular distribution data

We shall call this file search.in (to conform with the previous command file)

```
'om.fri' 'om.fro' 6 1
&variable kind=1 name='V' step=0.1 kp=1 pline=2 col=1 /
&variable kind=1 name='r' step=0.1 kp=1 pline=2 col=2 /
&variable kind=1 name='a' step=0.1 kp=1 pline=2 col=3 /
&variable kind=1 name='W' step=0.1 kp=1 pline=2 col=4 /
&variable kind=1 name='rw' step=0.1 kp=1 pline=2 col=5 /
&variable kind=1 name='aw' step=0.1 kp=1 pline=2 col=6 /
&data type=0 delta=0 iscale=2 idir=2 lab=F abserr=T/
     10.88  0.76100E+03 0.10654E+02
     13.05  0.75900E+03 0.11385E+02
     15.22  0.70900E+03 0.10635E+02
     17.39  0.62600E+03 0.93900E+01
     19.56  0.53600E+03 0.80400E+01
     21.72  0.44400E+03 0.66600E+01
     23.89  0.36500E+03 0.58400E+01
     26.05  0.27700E+03 0.41550E+01
```

To run SFRESCO, the command is very simple:

prompt:sfresco<sf.inp

The commands in the sf.inp file control the use of MINUIT to perform the χ^2 minimisation and the main output defaults to the screen (stream 6)

The sample file searches on 6 variables simultaneously – not usually recommended – but any number can be searched on

Constraints may be placed on variables individually in the form of limits which the search must remain within

Constraints may be placed on a particular variable or variables in the following way:

```
'om.fri' 'om.fro' 6 1
&variable kind=1 name='V' step=0.1 kp=1 pline=2 col=1 /
&variable kind=1 name='r' step=0.1 kp=1 pline=2 col=2 valmin=1.0 valmax=1.5/
&variable kind=1 name='a' step=0.1 kp=1 pline=2 col=3 /
&variable kind=1 name='W' step=0.1 kp=1 pline=2 col=4 /
&variable kind=1 name='rw' step=0.1 kp=1 pline=2 col=5 valmin=1.0 valmax=1.5/
&variable kind=1 name='aw' step=0.1 kp=1 pline=2 col=6 /
&data type=0 delta=0 iscale=2 idir=2 lab=F abserr=T/
     10.88  0.76100E+03 0.10654E+02
     13.05  0.75900E+03 0.11385E+02
     15.22  0.70900E+03 0.10635E+02
     17.39  0.62600E+03 0.93900E+01
```

Having (very briefly) covered optical model calculations and searches we shall now proceed to look at coupled channels (CC) calculations for inelastic scattering.

Since time is limited, we shall confine ourselves to collective states described using standard deformed potential models. FRESCO contains rotational model couplings as a standard procedure, thus doing much of the work for us, although finer control is allowed, thus enabling reorientation couplings to be omitted (if the strict vibrational model is followed) or “model independent” strengths to be used.

However, FRESCO does not, to the best of my knowledge, allow multi-phonon couplings as per the strict vibrational model.

We shall start with the simplest case, a strict rotational model calculation for the excitation of a single 2^+ level of an even-even nucleus. In order to illustrate the inclusion of Coulomb excitation we shall begin with a heavy-ion projectile this time, ^{7}Li, incident on a ^{12}C target.

We begin with the simplest possible input file, quadrupole excitation of the 4.44 MeV 2^+ level of ^{12}C. The only “complication” is that this is an oblate deformation, so that according to the conventions used in FRESCO we have to include negative signs.

What does the input look like, and how was it constructed?

```
** 7Li + 12C at 34.0 MeV
0.08    25.0    0.08
 00. 50.  +.00   F F
3 00.0     180.0  0.5  1
0.0    0   0 2 0  48
 1 1 0 0 1 3 0 0 0 1 0 0 1
7Li     7.0160  3.0         2   12C    12.000
1.5   -1 0.0000             1  0.0   +1 0.00
     1                      1  2.0   +1 4.438

  1 0  0    12.0     7.0    1.25
  111 10          -6.3008
  1 1  0   147.0    0.63    0.73     6.6
  111 11           -1.40
0

   0   1   1
34.0
EOF
```

Card 3
Card 4
Card 6
Card 7
Card 10
Card 10

Yes, it really is that simple: card types labelled in black you have seen before; card types labelled in red are new

These cards give the information about the coupling strengths, Coulomb first then nuclear. They also define how the couplings are implemented by FRESCO

Note the use of COPYp on Card 7; this is important

Taking the Coulomb excitation first, follow section 2.4 of the manual:

111 10 -6.3008

Card type 10:

KP=1 (index of potential);

TYPE=11, denoted in the manual as “Deformed target (matrix elements from ROTOR)”;

IT not used here, so left blank;

SHAPE=10. Only allowed value of SHAPE in this context. Denotes standard derivative based form factor for Coulomb excitation.

Finally, we have P2 = *Mn(E2)* = -6.3008 e fm^2.

All very well, but what exactly does “Deformed target (matrix elements from ROTOR)” mean and what is *Mn(E2)* and how do we choose its value for a particular case?

Taking the TYPE=11 first. This tells FRESCO how the inelastic coupling is to be performed. In this case, by deforming the preceding card 10 which defines the diagonal Coulomb potential and using an internal routine to calculate automatically all the allowed couplings according to the rotational model corresponding to the multipolarities and strengths supplied by the user.

Here we specify just multipolarity k=2, so only quadrupole couplings will be implemented.

How do we obtain the value of *Mn(E2)*? This is explained in section 2.4 of the FRESCO manual under *Deformations*.

The *Mn(Ek)*, where k is the multipolarity (2 here) are the so-called intrinsic reduced matrix elements. They are related to the reduced matrix elements *M(Ek)*, defined as:

$$M(Ek) = \pm\sqrt{(2I+1)B(Ek, I \to I')}$$

in the following way:

$$M(Ek) = Mn(Ek)(-1)^{[I-I'+|I-I'|]/2}\sqrt{2I+1}\,\langle IKk0 \mid I'K\rangle$$

This looks daunting at first sight but needn't be. We take it in stages and first obtain *M(Ek)*. This we do using the relationship:

$$M(Ek) = \pm\sqrt{(2I+1)B(Ek, I \rightarrow I')}$$

where *B(Ek,*$I \rightarrow I'$*)* is the usual *B(E2)*↑, frequently tabulated in the literature. A convenient source for even-even nuclei is: S. Raman *et al.*, Atomic Data and Nuclear Data Tables **78**, 1 (2001); *k* is just the multipolarity, so *k*=2 here.

I is the intrinsic spin of the initial state (ground state of ^{12}C here, so $I=0$) and I' the spin of the final state (4.44 MeV 2^+ of ^{12}C here, so $I' = 2$), and we have the simple result that in our case:

$$M(E2) = -\sqrt{B(E2, 0 \rightarrow 2)}$$

Points to note:

We chose the negative sign since this is the convention in FRESCO for oblate deformation

The *B(E2)* value must be in e^2 fm^4; often *B(E2)*s are tabulated in units of e^2 b^2 and will thus need to be multiplied by a factor of 10 000 to put them in the correct units

Make sure the value is for *B(E2)*↑, not *B(E2)*↓ as in most gamma-ray spectroscopy papers (the two values differ by a spin factor)

Now we have a value for *M(E2)* it is straightforward to obtain the desired *Mn(E2)*. The quantity:

<IKk0|I′K>

is the Clebsch-Gordan coefficient linking these parameters. The only one we have not so far defined is K, the K quantum number of the rotational band, often (but not always …) equal to the spin of the band head.

In this case, K=0, k=2, I=0 and I′=2 and we need the value of the following Clebsch-Gordon coefficient: <0020|20> = 1.0

A convenient on-line Clebsch-Gordon calculator may be found at:

https://www.volya.net/index.php?id=vc

Putting all this information into the relation:

$$M(Ek) = Mn(Ek)(-1)^{[I-I'+|I-I'|]/2}\sqrt{2I+1}\,\langle IKk0 \mid I'K\rangle$$

we obtain the result for our case of interest that:

$$M(Ek) = Mn(Ek)$$

Since the value of *B(E2; $0^+ \rightarrow 2^+$)* is listed in Raman *et al.* as 0.00397 e^2 b^2 we obtain the result that *M(E2)* = -6.3008 e fm^2.

This is a particularly simple case and things are more complicated if $I \neq 0$, but if you follow the above steps carefully you should have no real difficulty

Turning to the nuclear part now, and following section 2.4 of the manual:

111 11 -1.40

Card type 10:

KP=1 (index of potential);

TYPE=11, denoted in the manual as “Deformed target (matrix elements from ROTOR)”;

IT not used here, so left blank;

SHAPE=11: deformation of nuclear potential calculated by Gaussian quadrature. More accurate than the “standard” derivative method, especially for strong couplings.

Finally, we have P2 = DEF(2) = -1.40 fm, the nuclear deformation length, δ. This is related to the deformation parameter β by: δ = β R, where R is by convention the radius of the diagonal nuclear potential.

In this example we have the same δ value for both real and imaginary parts of the nuclear potential but different β values. This is based on the observation that δ is better determined than β, although they are both model dependent.

If you want to keep β the same for both real and imaginary parts of the potential you must replace the 2 lines in the example by the following:

```
1 1   0    147.0      0.63     0.73
111 11              -1.40
1 1   0                                  6.6     1.38     0.85
111 11           -3.0667
```

How did we arrive at the value of -1.40 fm for DEF(2)? The negative sign is easy: DEF(k) always has the same sign as *Mn(Ek)* in the Coulomb excitation part.

Where did the 1.40 fm come from? Since the nuclear deformation length is model dependent, ideally it should come from fitting inelastic scattering data for the specific case of interest, in this case for excitation of the 4.44 MeV 2^+ state of ^{12}C by a 34 MeV ^{7}Li beam. That is what was done here (in fact, the value was taken from a paper where this procedure was followed)

What does one do if such data are not available?

Try to find a similar analysis in the literature and use their value, or extract a Coulomb deformation length from the *Mn(Ek)* and assume the nuclear one is the same.

The FRESCO manual gives the relationship between *Mn(Ek)* and β. One also needs to assume a charge radius, and this is usually taken as $1.2 \times A^{1/3}$ fm. For ^{12}C this gives a value of R = 2.7473 fm.

Thus substituting in:

$$Mn(Ek) = 3\, Z\, \beta_k\, R^k/(4\pi)$$

we obtain: $-6.3008 = 3 \times 6 \times \beta_2 \times 2.7473^2/(4\pi)$,

so that $\beta_2 = -0.5828$ and $\delta = -1.60$ fm

This exercise usually gives values of δ that are somewhat too large, so if you take around 80% of the value arrived at it should be more reasonable.

Running an input that includes inelastic scattering coupling is of course exactly the same as for an optical model calculation, although it will take a little longer to finish!

However, depending on how strong the coupling is, you may need to re-fit the elastic scattering data if you are using optical potential parameters obtained from an optical model fit. This may be done with SFRESCO.

The comments concerning choice of “control” variables such as HCM, RMATCH etc. for optical model calculations still apply, although there is a more detailed way of choosing JTMAX (and therefore RMATCH) when performing inelastic scattering calculations. RINTP is also not used in CC calculations, so you may leave it the same as HCM.

At the very end of the main FRESCO output file some useful information is recorded:

```
ACCURACY ANALYSIS at    34.000 MeV :

 Elastic h*k = 0.171 so OK       compared with 0.200

  Real(S-el) > 0.01 first at J =       1.5
  Real(S-el) > 0.10 first at J =       8.5
  Real(S-el) > 0.50 first at J =      11.5
  Real(S-el) > 0.90 first at J =      15.5
  Real(S-el) > 0.99 first at J =      19.5
  R-turn =    24.30 fm    at J =      50.5

Forward-angle excitation cut off below   2.893 deg by max JT =      50.5
                               and below   2.773 deg by max R.

Total CPU  0 time =       0.41 seconds
```

It is the information under “Forward-angle excitation cut off below 2.893 deg by max JT = 50.5” that concerns us here.

This means that with JTMAX = 50, i.e. with a maximum of 50 partial waves included in the calculation, Coulomb excitation will be cut off for centre of mass scattering angles smaller than 2.893°.

In other words, the calculated inelastic scattering cross section is only accurate for angles greater than this value. If you require an accurate cross section at smaller angles you will need to increase JTMAX (and RMATCH).

This information allows the value of JTMAX to be set more precisely. You will need to decide at what point to stop, since attempting to reduce the cut-off angle to very small values will require an enormous number of partial waves (and a correspondingly large value of RMATCH).

Thus far we have examined a very simple case, coupling to a single state. How do we include other members of a (possibly hypothetical) rotational band?

If we assume for the purposes of the exercise that the 14.079 MeV 4^+ excited state of ^{12}C is the next member of a rotational band containing the 0^+ ground state and 4.44 MeV 2^+ excited state how do we add in coupling to this state within the rotational model?

If we further assume that ^{12}C has no hexadecapole deformation, i.e. that there is no k=4 coupling, nothing could be simpler:

```
** 7Li + 12C at 34.0 MeV
0.08     25.0     0.08
 00. 50.   +.00    F F
3 00.0      180.0   0.5   1
0.0     0    0 3 0   48
 1 1 0 0 1 3 0 0 0 1 0 0 1
7Li      7.0160   3.0          3    12C     12.0000 6.0       0.0
1.5    -1 0.0000               1   0.0    +1 0.00               F F
      1                        1   2.0    +1 4.4389
      1                        1   4.0    +1 14.079

  1 0   0     12.0      7.0     1.25
  111 10           -6.3008
  1 1   0    147.0     0.63     0.73      6.6     1.38     0.85
  111 11             -1.40
0

   0    1    1
34.0
EOF
```

That really is all there is to it! FRESCO's sub-routine ROTOR will automatically implement all allowed quadrupole couplings within this K=0 band with strengths defined according to the rotational model in accordance with the given *Mn(E2)* value.

Note that in this example we have used the actual excitation energy of the 4^+ state rather than the strict rotational model value, a compromise that is often used in calculations of this type.

If the nucleus you are interested in also has a static hexadecapole (k=4) deformation and you have a value for a *B(E4)* for direct excitation of the 4^+ level then you may include the allowed k=4 couplings by adding the appropriate *Mn(E4)* value:

```
** 7Li + 12C at 34.0 MeV
0.08     25.0     0.08
 00. 50.   +.00    F F
3 00.0      180.0   0.5   1
0.0     0    0 3 0   48
 1 1 0 0 1 3 0 0 0 1 0 0 1
7Li      7.0160  3.0          3   12C     12.0000 6.0      0.0
1.5    -1 0.0000              1  0.0    +1 0.00              F F
     1                        1  2.0    +1 4.4389
     1                        1  4.0    +1 14.079

  1 0  0     12.0      7.0     1.25
  111 10           -6.3008         XXXXXXXX
  1 1  0    147.0     0.63     0.73      6.6     1.38     0.85
  111 11             -1.40         xxxxxxxx
0

   0    1    1
34.0
EOF
```

This is all very well if we only require coupling according to the rotational model. What if we wish to consider vibrational levels or if we have experimentally determined strengths for the different couplings (for example, re-orientation couplings)?

FRESCO can also handle this, by using potential TYPE=12 or 13 for projectile and target coupling respectively. Using these TYPEs it is possible to specify individually the various coupling strengths, independent of any model.

Implementation is slightly more complicated than with TYPEs 10 and 11, so we shall simply repeat the rotational model calculations inserting the coupling strengths “by hand” as an example.

```
** 7Li + 12C at 34.0 MeV
0.08     25.0     0.08
 00. 50.   +.00    F F
3 00.0      180.0   0.5   1
0.0     0    0 2 0   48
 1 1 0 0 1 3 0 0 0 1 0 0 1
7Li      7.0160   3.0          2    12C     12.0000 6.0      0.0
1.5    -1 0.0000               1   0.0    +1 0.00              F F
     1                         1   2.0    +1 4.4389

  1 0  0     12.0      7.0     1.25
  113 10             -6.3008
Card 11           1    2    2 -6.3008
Card 11           2    1    2 -6.3008
Card 11          -2    2    2  7.5309
  1 1  0    147.0     0.63     0.73      6.6     1.38     0.85
  113 11             -1.40
Card 11           1    2    2   -1.40
Card 11           2    1    2   -1.40
Card 11          -2    2    2  1.6733
0

   0    1    1
34.0
EOF
```

The difference from the previous input is the presence of Cards of type 11. These specify the strengths for each of the allowed transitions individually.

How is this done?

According to the manual, Cards 11 are of the following format:

```
1    2    2 -6.3008
```

this particular line representing the coupling from state 2 to state 1 of multipolarity k=2 and strength STR=-6.3008. This example is for Coulomb excitation, so STR = *M(Ek)* which, for this particular transition, is equal to *Mn(E2)*. To see how this result is obtained, we go back to the relation:

$$M(Ek) = Mn(Ek)(-1)^{[I-I'+|I-I'|]/2}\sqrt{2I+1}\,\langle IKk0 \mid I'K\rangle$$

We know *Mn(E2)* = -6.3008 e fm^2 and we need to find *M(E2)* for the $2^+ \rightarrow 0^+$ transition. Substituting all the quantities we find:

M(E2) = -6.3008 $(-1)^2$ $\sqrt{5}$ <2020|00> and since <2020|00> = $1/\sqrt{5}$ we have that *M(E2)* = -6.3008 e fm^2

The line that contains the re-orientation coupling for the 2^+ level, i.e.

-2 2 2 7.5309

has a different value for STR because *M(E2)* is now given by:

M(E2) = -6.3008 $(-1)^0$ √5 <2020|20> = 7.5309 since <2020|20> = -0.534522

N.B. there is no line for the reorientation of the 0^+ level since this is not allowed. The negative value at the beginning of the line merely indicates that this is the last Card type 11 to be read in at this point.

Similar considerations apply for the nuclear deformation, with STR = RDEF and *Mn(Ek)* replaced by DEF(k)

There is one final point to note. If you wish to include coupling to the ^{12}C 3^- level at 9.64 MeV as an octupole phonon type coupling in addition to the 2^+ level treated as a rotational state you must define the 3^- coupling strength separately, as follows:

```
** 7Li + 12C at 34.0 MeV
0.08    25.0    0.08
 00. 50.  +.00    F F
3 00.0     180.0  0.5  1
0.0    0   0 3 0  48
 1 1 0 0 1 3 0 0 0 1 0 0 1
7Li     7.0160  3.0        3   12C    12.0000 6.0     0.0
1.5   -1 0.0000            1  0.0   +1 0.00            F F
      1                    1  2.0   +1 4.4389
      1                    1  3.0   -1 9.641

  1 0  0    12.0     7.0    1.25
  111 10         -6.3008
  113 10                24.6982
       3   1   3 24.6982
      -1   3   3 24.6982
  1 1  0   147.0    0.63    0.73     6.6    1.38    0.85
  111 11           -1.40
  113 11                    1.0
       3   1   3 1.0
      -1   3   3 1.0
0

   0   1   1
34.0
EOF
```

If you do not do this and instead simply add the 3^- strengths to the Cards 10 with TYPE=11 coupling FRESCO will treat the 3^- level as part of the same K=0 band as the 2^+ level and introduce couplings between the 2^+ and 3^- levels you do not want!

You should now know the basics of how to perform coupled channels calculations with FRESCO. The practical exercises tomorrow will take you through a number of other examples including the slightly more complicated cases where the ground state spin is not zero.

I expect you are all more than ready for your coffee break! When we reconvene we will look at modelling transfer reactions with FRESCO within the DWBA and CRC formalisms.

Thank you for your attention

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